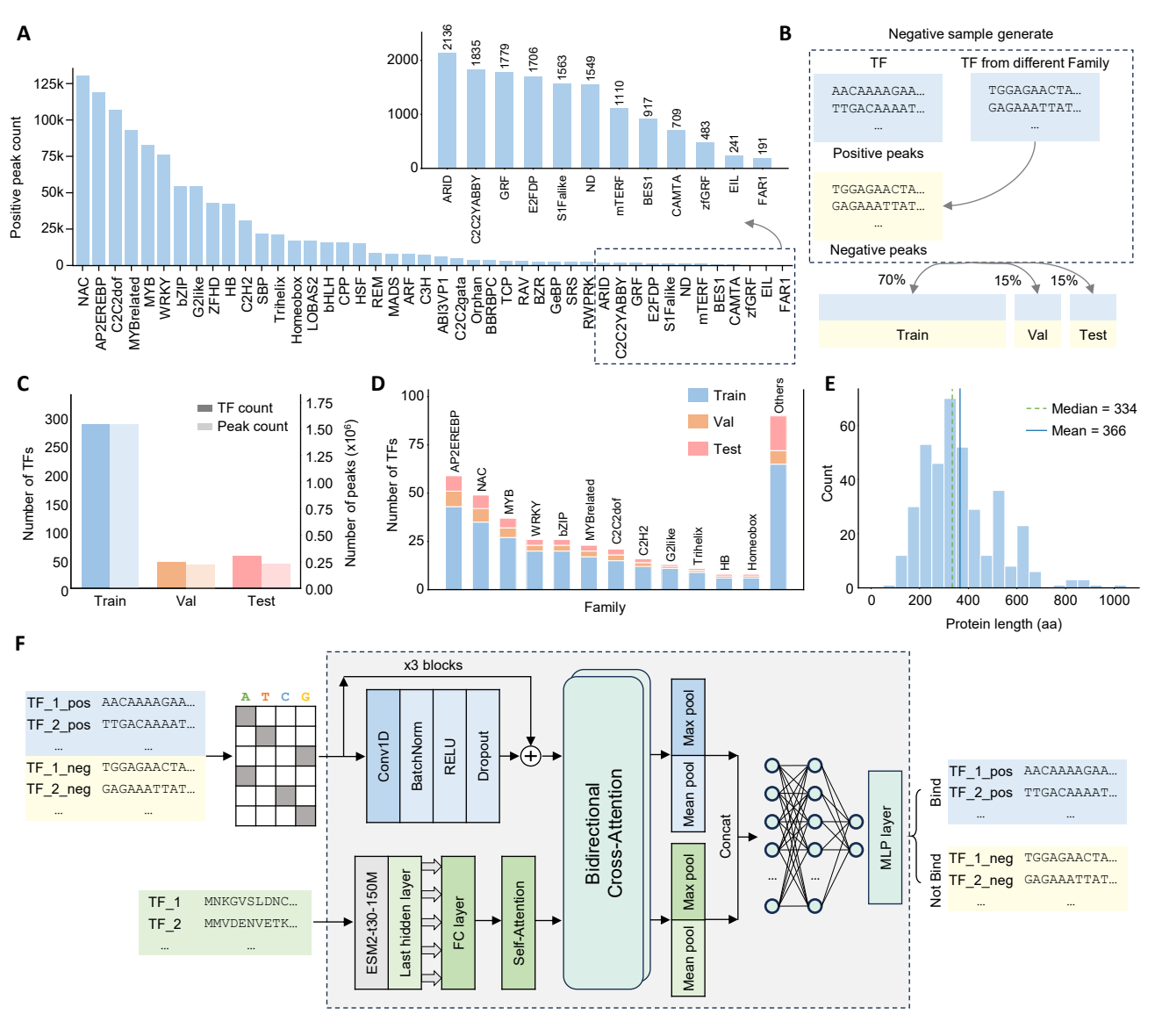

A
Positive peak count
125k
100k
75k
50k
25k
0
NAC
AP2EREBP
C2C2dof
MYBrelated
MYB
WRKY
bZIP
G2like
ZFHD
HB
C2H2
SBP
Trihelix
Homeobox
LOBAS2
bHLH
CPP
HSF
REM
MADS
ARF
C3H
ABI3VP1
C2C2gata
Orphan
BBRBPC
TCP
RAV
BZR
GeBP
SRS
RWPRK
ARID
C2C2YABBY
GRF
E2FDP
S1Falike
ND
mTERF
BES1
CAMTA
zfGRF
EIL
FAR1
2000
1000
0
2136
1835
1779
1706
1563
1549
1110
917
709
483
241
191
ARID
C2C2YABBY
GRF
E2FDP
S1Falike
ND
mTERF
BES1
CAMTA
zfGRF
EIL
FAR1
B
Negative sample generate
TF
TF from different Family
AACAAAAGAA...
TTGACAAAAT...
...
TGGAGAACTA...
GAGAAATTAT...
...
Positive peaks
TGGAGAACTA...
GAGAAATTAT...
...
Negative peaks
70%
15%
15%
Train
Val
Test
C
TF count
Peak count
Number of TFs
Number of peaks (x10^6)
300
250
200
150
100
50
0
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Train
Val
Test
D
Train
Val
Test
Number of TFs
100
75
50
25
0
AP2EREBP
NAC
MYB
WRKY
bZIP
MYBrelated
C2C2dof
C2H2
G2like
Trihelix
HB
Homeobox
Others
Family
E
Median = 334
Mean = 366
Count
60
40
20
0
0
200
400
600
800
1000
Protein length (aa)
F
TF_1_pos AACAAAAGAA...
TF_2_pos TTGACAAAAT...
... ...
TF_1_neg TGGAGAACTA...
TF_2_neg GAGAAATTAT...
... ...
A T C G
x3 blocks
Conv1D
BatchNorm
RELU
Dropout
TF_1 MNKGVSLDNC...
TF_2 MMVDENVETK...
... ...
ESM2-t30-150M
Last hidden layer
FC layer
Self-Attention
Bidirectional Cross-Attention
Max pool
Mean pool
Max pool
Mean pool
Concat
MLP layer
Bind
Not Bind
TF_1_pos AACAAAAGAA...
TF_2_pos TTGACAAAAT...
... ...
TF_1_neg TGGAGAACTA...
TF_2_neg GAGAAATTAT...
... ...